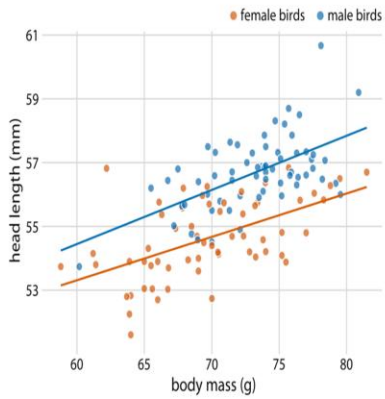
Fitting Curves with a Functional Form to Data

- preferable to smoothing techniques because they are more predictable
- provide parameter estimates with meaningful interpretations

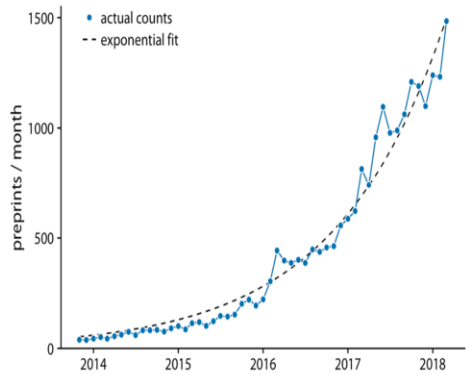
The Linear Form:

$y = mx + b$



A Non-Linear Example

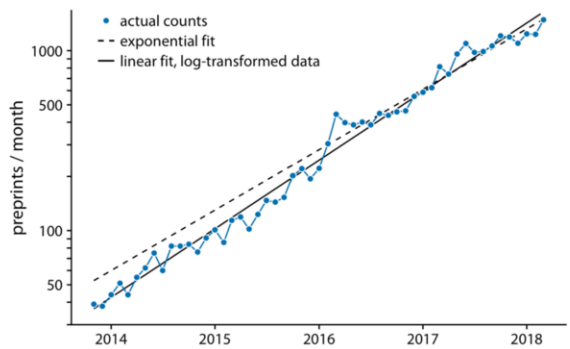
- increase in submissions in each month is proportional to the number of submissions in the previous month



Checking for an Exponential Relationship

Take the log of the y values (or equivalently, perform a log scaling of the y-axis) and plot, looking for a linear relationship

Log-Linear Plot



- Blue line : linear fit after transformation
- Dotted line : exponential fit after transformation
    - Despite data is transformed algorithm for different fits are different
    - Exponential fit applied after transformation, but calculations are different
        - Therefore despite it being a transformed exponential graph a linear fit would work better then an exponential fit

For Non-linear Relationships…

It is usually better to fit a straight line to transformed data than to fit a nonlinear curve to untransformed data.